Information Security

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Motivation


اَلْحَمْدُ لِلّٰهِ رَبِّ الْعٰلَمِيْنَ ۙ (۱)
سب خوبیاں اللہ کو جو مالک سارے جہان والوں کا

Examples of Security Requirements: Confidentiality

- Student grade information is an asset whose confidentiality is considered to be very high.
  - The **US FERPA Act**: grades should only be available to students, their parents, and their employers (when required for the job)
- Student enrollment information: may have moderate confidentiality rating; less damage if enclosed
- Directory information: low confidentiality rating; often available publicly

Examples of Security Requirements: Integrity

- A hospital patient's allergy information (high integrity data): a doctor should be able to trust that the info is correct and current
  - If a nurse deliberately falsifies the data, the database should be restored to a trusted basis and the falsified information traced back to the person who did it
- An online newsgroup registration data: moderate level of integrity
- An example of low integrity requirement: anonymous online poll (inaccuracy is well understood)

Examples of Security Requirements: Availability

- A system that provides authentication: high availability requirement
  - If customers cannot access resources, the loss of services could result in financial loss
- A public website for a university: a moderate availably requirement; not critical but causes embarrassment
- An online telephone directory lookup: a low availability requirement because unavailability is mostly annoyance (there are alternative sources)

Challenges Of Computer Security

- Computer security is not simple
- One must consider potential (unexpected) attacks
- Procedures used are often counter-intuitive
- Must decide where to deploy mechanisms
- Involve algorithms and secret info (keys)
- A battle of wits between attacker/admin
- It is not perceived as a benefit until fails
- Requires constant monitoring

Jazak Allah Khair!